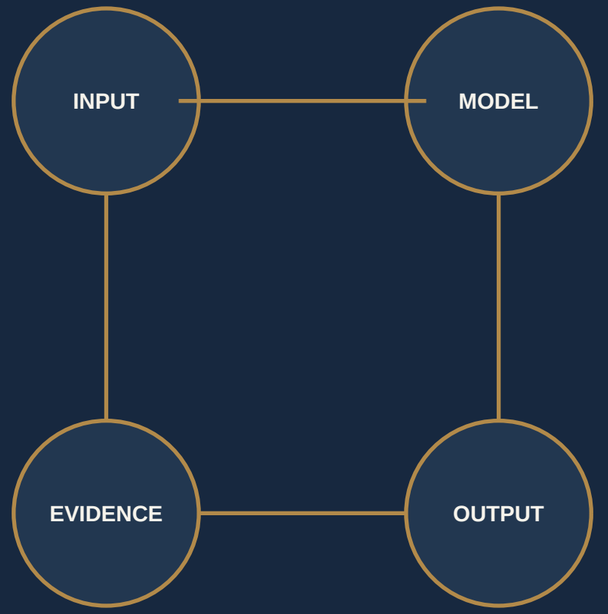
BACHELOR OF ARTIFICIAL INTELLIGENCE & ROBOTICS ENGINEERING

Data Structures and Algorithms

AI.107 | University lecture series



What you will learn

A practical progression from core ideas to engineering judgment

- Explain and define arrays.
- Apply key methods including linked lists.
- Build or analyze a practical system involving stacks.
- Evaluate assumptions, limitations, performance and responsible use.

A connected learning journey

Each stage prepares the tools needed for the next.

01

FOUNDATIONS

Arrays

02

METHODS

Linked lists

03

SYSTEMS

Stacks

04

PRACTICE

Laboratory, evaluation and project

The subject at a glance

A map of the ideas and techniques in this course

01 · Arrays
Arrays

02 · Linked lists
Linked lists

03 · Stacks
Stacks

04 · Queues
Queues

05 · Trees
Trees

06 · Graphs
Graphs

07 · Hash tables
Hash tables

08 · Heaps
Heaps

Core ideas, made explicit

AI.107 • Distinguish definitions, assumptions and evidence

- Arrays
- Linked lists
- Stacks
- Queues
- Trees

DEFINE → MODEL → TEST → INTERPRET

Core ideas, made explicit

AI.107 • Distinguish definitions, assumptions and evidence

- Graphs
- Hash tables
- Heaps
- Recursion
- Sorting

DEFINE → MODEL → TEST → INTERPRET

Follow the system through its stages

Use the process diagram to connect components and outcomes.

PROBLEM
Arrays

INPUTS
Linked lists

METHOD
Stacks

OUTPUT
Queues

EVALUATION
Evidence, constraints and limitations

Teaching sequence: reveal one component at a time; then replay the complete end-to-end flow.

From concept to a real engineering decision

A short case study for discussion and worked-example narration

SCENARIO

A real task involving arrays.

CONSTRAINTS

Data, resources, safety and operating conditions shape the design.

DECISION

Select a method; justify the trade-offs and define success.

- State assumptions before calculating or implementing.
- Compare the expected result with observed evidence.

Laboratory · build, measure, improve

Students work through the following guided sequence.

- Specify a measurable task using arrays.
- Implement or simulate a baseline using linked lists.
- Run a controlled test and record results with units and conditions.
- Diagnose a failure, revise the design and explain the improvement.

Review · explain · apply

KNOWLEDGE CHECK

What assumptions does the method make? What evidence would show that it works?

PRACTICAL CHALLENGE

Apply one course idea to a new context; state constraints, test criteria and likely failure modes.

Key takeaway: make the model visible, test it, and communicate its limits.

Course-series slide design • detailed narration, demonstrations and assessments expand each module.